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The Maritime Industry: Backbone of the Global Economy

Maritime is defined as "of or relating to navigation or commerce on the sea," according to the Webster dictionary. However, this barely scratches the surface of what Maritime is. Personally, I view Maritime as an industry that significantly benefits everyone from the individual to entire nations. Some of the more significant benefits are seen economically, as many economies from the local to the international level are dependent on the maritime industry, or societally, as the industry can shape the culture in multiple ways. That being said, the industry requires skilled workers in conjunction with technological innovations to remain effective and efficient. It's also important to keep the environment in mind as the industry negatively affects it but is actively working to become more sustainable and cleaner.

Since humans began to abandon the hunter-gatherer lifestyle and form societies, populations have found themselves focused heavily on where there is water due to the immense economic benefits, which led to these societies being shaped by their reliance on the maritime industry. Not only does it influence trade and create jobs, but it also leads to infrastructure development. Trade is the shining example of maritime influence, as 80% of tangible products are shipped by sea. (World Bank Group). The two biggest reasons for this are that shipping goods by sea is the most cost-effective method of international shipping, and globalization is increasing as economies become more dependent on each other for certain products (depending on factors such as comparative/absolute advantage). Furthermore, due to the relatively low cost of ocean shipping, this allows international companies to provide consumers with a much wider variety of products. A relevant example of this is that the top supplier of goods to the U.S. is China, with 16.5% as of 2022. (“Countries & Regions | United States Trade Representative”) Due to the low cost of shipping goods by sea, the American consumer is able to reasonably take advantage of the low cost of goods as a result of low Chinese labor costs. Job creation is an occurrence synonymous with the maritime industry. According to the National Oceanic and Atmospheric Administration (NOAA), the U.S. Marine economy is responsible for employing 2.6 million Americans annually. Furthermore, this job growth indirectly stimulates job growth itself by attracting more businesses and regional investment, along with all who support those, and inspiring further regional/local economic development. Infrastructure development is also a fruit of maritime economies. Shipyards and ports always require some sort of intermodal infrastructure, such as roads for trucks and railroads for trains, to get goods from the ship they arrived on to their final destination (and vice versa). Furthermore, these locations require heavy machinery and equipment such as material handlers (cranes, fork trucks, tractors, etc.) Warehouses are also needed for storage between cargo movements. Ultimately, all of this infrastructure creates logistic hubs, which attract further revenue streams (directly & indirectly) into the area. Overall, the maritime industry greatly contributes to the economy and societal development by influencing trade, creating jobs, and also developing infrastructure that can add economic benefits itself.

The maritime industry would be nothing without its workers, who support supply chains all around the world, which is ultimately a critical part of the global economy. Whether a longshoreman or the operations manager at a port, there are certain skills that are needed. One of the most important skills is simply flexibility. Just like any industry in the past few decades, the maritime industry has experienced rapid growth, change, and development in general. That being said, the ability to adapt to changes (whether in the internal or external environment) can result in much more experienced and well-rounded workers. Furthermore, these workers are nothing without their leadership, who are responsible for decision-making of varying degrees. One of the leadership traits that is integral for the maritime field is judgement. Although a fairly cliché trait, leaders in general need to be able to make decisions with calmness, evaluating all evidence, and do such promptly. This is simply because they need to not only respond adequately but also make timely decisions in stressful situations. After all, history has shown us the importance of quick decision-making. A great example of this was during the Port of Houston cyber attack in 2021, where port leadership quickly enacted its facilities security plan and thwarted the attack. Due to their outstanding judgment, they were able to prevent a major supply chain disruption that would have devastating effects on the economy from the local to the national level. Ultimately, the workers and leaders are the backbone of the maritime industry, a critical part of the global economy.

The maritime industry is just like any other business, taking advantage of technologies to become leaner, optimize processes, and increase throughput of cargo. A technological innovation that would improve the Hampton Roads maritime industry would be increasing automation at the Port of Virginia. Although there is some automation already, automating operations would reduce labor costs, increase efficiency, and mitigate human errors. This can be seen in the Port of Rotterdam, where fully autonomous operations have led to “production increases of 80%” (“The Importance of Port Automation in America”). Although it’s difficult to put a price tag on this hypothetical upgrade, it would likely cost around $4.5 billion to go full automation and take 10-20 years, based on other fully automated port projects. However, the Port of Virginia has already made some automated implementations, which could make it cheaper and for a much smoother transition. Unfortunately, greater efficiency comes at a cost. One of those is the reduction of jobs. However, the Port of Virginia has ironically had an issue with this more recently, as longshoremen have gone on strike to significantly increase pay and deter further automation. This increased technological utilization would also lead to greater energy costs due to more energy being required to sustain the new automation infrastructure. Furthermore, automation requires a significant amount of data collection and processing, which would require data warehouses that are extremely costly in all aspects. On the bright side, this would attract more business to the Port and most likely reduce costs enough to compensate for other increased costs. Overall, increased automation of the Port of Virginia would significantly impact the Hampton Roads region both positively and negatively.

The maritime industry greatly benefits many; however, it doesn’t benefit the environment. Although it’s been empirically proven that ocean shipping is the least damaging mode of transportation, which is especially important considering that it facilitates 90% of global trade, there is still great responsibility and accountability needed for the rest of the industry concerning environmental care. These responsibilities extend to ports as well, such as the Port of Virginia, which in 2024 was able to finally power all terminals from 100% clean energy sources. Furthermore, they plan on achieving net-zero carbon emissions by the year 2040. (“Environmental Sustainability - Port of Virginia”) Although environmental sustainability has become more popular in the 21st century, there are still many issues, such as oil spills, which can affect marine life by poisoning, altering their physical attributes, which help them survive, or even simply reducing their food supply and causing them to starve. (National Oceanic and

Atmospheric Administration). In the grand scheme of things, even the maritime industry is responsible for reducing its effects on the environment and practicing sustainability.

All things considered, to me, the maritime industry is an entity that greatly benefits the economy and society. It also provides a space for technological innovation and development, such as automation for efficiency improvements. None of these benefits would be possible without skilled workers and their talented leadership. Although it does have many benefits, it also has some drawbacks, such as environmental impacts; however, the industry continues to make active strides in counteracting damage made in the past. The maritime industry not only means a benefit to me, but also a great portion of society.

Personal Takeaways, Future Interests, & Course Enhancement Suggestions

The topic I found most insightful was automation in maritime shipping. Automation has become much more prevalent in today's business environment, as data is relied on more heavily than ever before in decision-making, with the ultimate objective being optimization to lower costs. Supply chain falls under this umbrella, as many supply chain functions, from route optimization to distribution center operations, have been fully or partially automated. Furthermore, this has extended to ocean shipping, which has found itself the next subject of automation. However, automation in this case doesn't mean lacking any human oversight or control, as semi-autonomy seems to be much more attainable for the time being, where certain functions, processes, or even systems that go on board a vessel are automated as opposed to the entire vessel being autonomous. Regardless, this will create great value to many shipping companies as they can create a much leaner cost structure by reducing skilled labor needed to man vessels. Furthermore, it would be a solution to the issue of the ever-decreasing number of skilled workers in the maritime industry, mitigating what could adversely affect the global supply chain in the future. Electronic Data Interchange (EDI) is something I'd be more interested in studying. The first reason is that I'm a believer in how data will continue to increase in importance and utilization for decision-making. On top of that, most ports utilize EDI for tracking shipments, logistics accuracy, and overall process optimization (to reduce queues and costs). I'm also a business analytics major and have been taught the value that data-driven insights can bring to industries, let alone businesses. That being said, EDI can serve as a collection point for data, giving it potential machine learning (ML) applications to provide those same data-driven insights.

The only additional topic I wish I had seen on this is underway operations. Although this wouldn't need to go into grand detail or exploration of very technical aspects, it would be nice to understand the life cycle of a ship's journey from launching at a port to its docking at a destination. The objective of this would be to give students a greater idea of the bigger picture of maritime-related supply chain activities.

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